

Revised Generation Interconnection System Impact Study Report

For Queue Project Q20-1516

Esc. Vocacional 138 kV
April 02, 2022

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q20-1516. The Interconnection Customer proposes a 188.7 MW battery energy storage facility interconnecting at the Esc. Vocacional 138 kV substation in the New Jersey area. The request was received on July 06, 2020. The study evaluated system intact and single contingency conditions using the regional planning case, including 437 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 1 overloaded facility. The total estimated cost of the network upgrades allocated to this project is \$5,181,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
ESC. VOCACIONAL to GOULDS 138 KV line	ESC. VOCACIONAL to CASTRO VLY TAP 138 KV	101.3	214	100.0

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
ESC. VOCACIONAL to GOULDS 138 KV	Reconductor line	\$5,181,000	100% (\$5,181,000)
	Total allocated to project		\$5,181,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.